Biophilic Design and Climate Change

Rethinking Mental Health infrastructure in a warming world

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Climate-responsive design typically focuses on flood walls, reflective roofing, and resilient infrastructure – the physical systems that help buildings withstand environmental stress. We track carbon footprints. We optimise passive cooling. We manage stormwater. All necessary. All insufficient.

What we're missing is more fundamental: climate change is rewiring how human brains manage stress, process information, and maintain emotional stability. And unlike a heat wave you can escape, chronic neurological stress compounds over time in ways that undermine the very cognitive and emotional capacities we need to adapt.

The evidence has been building for years, but it's rarely centred in discussions of climate adaptation. Systematic reviews show that higher ambient temperatures and temperature variability correlate with increased anxiety, depression, psychiatric hospitalizations, and suicide attempts across diverse geographic regions (Thompson et al., 2023; Rony et al., 2023). In July 2024, Phoenix hit 31 consecutive days above 110°F. Emergency departments reported a 40% surge in mental health crises alongside the expected heat-related illness.

Climate stress is as much a neurological threat as a physical one, and the built environment either amplifies or buffers that threat depending on how it's designed.

A note on evidence: While research linking biophilic design to mental health outcomes under climate stress is growing rapidly, many studies remain correlational rather than causal, and long-term intervention data are still emerging. This article synthesises current evidence while acknowledging gaps that should define future research priorities.

The Neurological Architecture of Climate Stress

Most climate adaptation frameworks treat human response as binary: you're either heat-stressed enough to require medical intervention, or you're fine. That misses the continuous neurological burden that accumulates below the threshold of clinical illness – the cognitive impairment, emotional dysregulation, and attention depletion that erode daily functioning long before anyone shows up in an emergency room.

The biological mechanisms are more direct than many designers realise. Heat exposure disrupts the hypothalamus, which governs both body temperature and the hypothalamic–pituitary–adrenal (HPA) axis – the body's primary stress response system. As temperatures rise, cortisol secretion increases, heart rate variability decreases, and parasympathetic recovery (the nervous system's ability to return to calm) is delayed (Baecker et al., 2025). These aren't subtle effects. They manifest as heightened anxiety, reduced emotional control, and a nervous system chronically biased toward threat detection.

Sleep disruption represents one of the most underappreciated pathways. Elevated nighttime temperatures interfere with slow-wave and REM sleep – phases essential for emotional processing and memory consolidation. Population studies show that even modest nighttime heat increases correlate with mood disturbance and next-day cognitive fatigue, particularly in urban areas where buildings and pavement retain heat (Thompson et al., 2023; Rony et al., 2023). The cumulative effect of poor sleep under sustained heat exposure compounds over weeks and months, eroding baseline emotional regulation in ways that don't resolve with a single cool night.

Cognitive function takes a direct hit as well. Working memory shrinks under thermal stress. Executive function declines. Decision-making gets slower and less accurate. The brain redirects metabolic resources away from thinking toward just maintaining body temperature. Sustained heat narrows attention, slows reaction time, and cranks up irritability – effects documented in both laboratory settings and field studies during heat waves (Baecker et al., 2025). These cognitive effects show up before clinical heat illness occurs, which means the brain registers and responds to thermal stress earlier than other body systems.



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Uncertainty as Neurological Stressor

Beyond direct heat exposure, climate change introduces something arguably more insidious: chronic environmental uncertainty. The brain evolved to anticipate and predict environmental conditions as a core survival mechanism. When environments become fundamentally unpredictable, they activate threat-detection systems even without immediate danger. Functional imaging studies show that uncertainty increases amygdala activation and disrupts prefrontal regulation, leading to heightened vigilance, anxiety, and cognitive fatigue (Peters et al., 2017).

Climate anxiety – persistent fear of environmental catastrophe – is no longer limited to activists or climate scientists. It's showing up across age groups and geographies, particularly among young people and those in climate-vulnerable regions (Clayton et al., 2017; Pihkala, 2020). When uncertainty becomes chronic, as it is with climate instability, it impairs emotional regulation, increases rumination, and erodes agency – the sense that your actions matter.

Why the Built Environment Matters

Buildings and landscapes aren't neutral. They either amplify or buffer these neurological effects depending on their material properties, spatial organization, and sensory qualities. Structures dominated by dark materials, minimal ventilation, and poor shading trap heat and compound thermal stress. Disordered landscapes and visually chaotic streetscapes increase cognitive load by forcing constant threat-scanning without reliable reference points (Evans, 2003). The cumulative effect of these design failures is measurable: higher baseline stress, faster attention depletion, and reduced capacity for emotional regulation.

Conversely, environments designed for thermal moderation, spatial legibility, and sensory coherence function as neurological buffers. They reduce stress intensity, enable faster recovery, and preserve cognitive capacity during extreme weather. This isn't speculative – it's documented in comparative studies of neighbourhoods with and without green infrastructure, before-and-after analyses of biophilic interventions, and controlled experiments examining physiological responses.

Biophilic design operates through multiple overlapping mechanisms: physical cooling via vegetation and shade, perceptual effects influencing how the brain appraises environmental safety, and spatial organization that supports rather than taxes cognitive function. These mechanisms work together, not in isolation. Interventions addressing only one dimension (say, technical cooling without spatial legibility) consistently underperform compared to integrated approaches.

The conventional framing treats biophilic design as aesthetic enhancement or environmental amenity. That fundamentally misunderstands what's happening neurologically.

In a climate-stressed environment, biophilic design becomes mental health infrastructure – as essential as cooling centres or emergency medical systems but operating preventively rather than reactively.

Design Strategies for Neurological Resilience

If we accept that climate stress is fundamentally neurological, the design challenge shifts from "how do we cool buildings" to "how do we create environments that support nervous system regulation under chronic stress." The distinction matters because it changes what we prioritise, how we measure success, and where we intervene.

Physical Cooling: Necessary But Not Sufficient

Urban green infrastructure reduces ambient temperatures through evapotranspiration and shade. The effects are substantial: 1-5°C reductions depending on vegetation density and spatial configuration (Aram et al., 2019; Koc et al., 2018). Even small temperature decreases produce measurable improvements in thermal comfort and reductions in physiological stress.

Singapore's approach illustrates both the potential and the limitations of focusing primarily on physical cooling. The city's Park Connector Network – over 300 kilometres of shaded, vegetated pathways linking neighbourhoods to parks – was designed mainly for active transportation and recreation. Post-implementation studies documented 2-4°C temperature reductions along corridors and 35% increases in pedestrian and cycling activity. But researchers were surprised by the magnitude of self-reported stress reduction among regular users, which exceeded predictions based on temperature reduction alone.

The explanation: temperature reduction matters, but so does perceived refuge, spatial legibility, and sensory quality. The Park Connectors work neurologically because they provide graduated transitions between different thermal zones, clear wayfinding, and consistent yet varied natural elements – addressing multiple stress pathways simultaneously.

Perceptual Effects: The Underestimated Mechanism

Most designers understand that vegetation cools through shade and evapotranspiration. Fewer recognise that environments perceived as cooler, greener, or more natural reduce subjective heat stress and improve mood even when objective temperature reductions are modest (Yao et al., 2021). This isn't placebo – it's cognitive appraisal, the brain making rapid assessments about environmental safety and resource availability.



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The perceptual dimension matters because it influences behaviour and stress responses independently of measured temperature. Perceived thermal comfort shapes how long people stay in outdoor environments, whether they use active transportation, and how quickly their stress responses activate during heat exposure. Biodiverse planting schemes, water features, and spatial complexity all contribute to environmental appraisal in ways that amplify physical cooling.

This creates design leverage. Interventions addressing both physical and perceptual dimensions produce compound benefits: actual temperature reduction plus enhanced environmental appraisal, resulting in lower physiological stress and improved emotional regulation.

Spatial Organization: Networks Over Nodes

Many cities treat thermal refuge as discrete nodes – a cool park here, a shaded plaza there – rather than continuous networks. Heat-related stress accumulates across daily movement patterns. If someone has to traverse 800 meters of exposed pavement between shaded areas, the cumulative thermal load undermines whatever relief those refuges provide.

Melbourne's Urban Forest Strategy recognised this early. Rather than just increasing overall tree canopy (though they set a 40% coverage target by 2040), they prioritised "cool corridors" linking transportation hubs to parks, schools, and commercial districts. The design principle was simple: no one should have to walk more than 200-300 meters between shaded rest opportunities during extreme heat.

Early evaluation (2019-2023) showed the strategy working – 2-3°C temperature reductions in established corridors, 28% reduction in heat-related emergency calls in targeted neighbourhoods, and increased pedestrian activity during extreme heat days. But the mental health outcomes were more striking: residents in corridor-adjacent neighbourhoods reported lower perceived stress and better emotional regulation during heat waves compared to demographically similar neighbourhoods without corridor access.

The network effect matters because it supports graduated thermal transitions rather than repeated exposure-relief-exposure cycles that tax the nervous system. This is particularly important for populations with limited mobility, cognitive impairments, or pre-existing mental health conditions – those disproportionately vulnerable to thermal stress.





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Temporal Layering: Immediate and Long-Term Strategies

One implementation challenge that rarely gets discussed openly: trees take time to mature, but people are heat-stressed now. Waiting 15-20 years for mature canopy cooling effects while residents suffer through increasingly intense heat waves is neither ethically defensible nor politically viable.

Successful projects layer interventions across temporal scales. Built shade structures – pergolas, shade sails, covered walkways, strategically designed building overhangs – provide immediate relief. Strategic tree species selection (including fast-growing species like London plane, zelkova, or regional equivalents adapted to local conditions) can yield substantial shade within 3-5 years (Aram et al., 2019). Mature canopy effects peak at 15-20 years. By combining approaches, projects deliver both immediate relief and compounding long-term benefits.

The temporal dimension also influences how communities perceive and engage with interventions. Visible, immediate improvements build political support and community trust, which sustains commitment through the longer timeline required for full ecological function.

Water: Cooling and Calming

Water features enhance microclimatic cooling through evaporative processes, but their psychological effects may be equally important. Research on natural elements including water shows these features correlate with reduced stress, lower blood pressure, and improved mood (Ulrich et al., 1991; Yao et al., 2021). The mechanism likely involves both direct sensory effects (sound and sight of water activating parasympathetic responses) and cognitive associations (water signalling resource availability and safety).

Fountains, misting systems, and water-adjacent landscapes reduce local air temperature while providing multisensory refuges that support stress recovery. When combined with shade and vegetation, water features create microclimates where both physical cooling and psychological restoration occur simultaneously.

Equity and the Distribution of Recovery

Climate stress exposure follows the geography of historical disinvestment with disturbing precision. Neighbourhoods shaped by redlining and discriminatory housing policy run significantly hotter due to reduced tree canopy, increased impervious surfaces, and lower-quality building stock (Hoffman et al., 2020). Temperature differences of 10-15°F within the same city on the same day, along lines mapping almost perfectly onto historical race-based lending restrictions, have been documented across major U.S. cities (Hsu et al., 2021).



Elevated neighbourhood temperatures correlate with increased heat-related illness and mortality, but also with higher rates of anxiety, mood disorders, and other stress-related mental health outcomes (Thompson et al., 2023). The chronic exposure compounds over time, increasing allostatic load – cumulative physiological wear – and eroding baseline emotional regulation (McEwen & Akil, 2020).

Green Space as Protective Factor, Unequally Distributed

Large-scale epidemiological studies show proximity to parks, street trees, and biodiverse landscapes associates with lower psychological distress, reduced antidepressant use, and improved self-reported mental health, even controlling for socioeconomic factors (Twohig-Bennett & Jones, 2018). These benefits are strongest in populations facing higher baseline stress, suggesting nature exposure operates as protective factor against cumulative adversity.

The problem is access. Communities with fewer green amenities often lack shaded streets, safe outdoor spaces, and cooling infrastructure, forcing residents to endure higher thermal and sensory stress during extreme weather (Hsu et al., 2021). This creates a vicious cycle: communities most exposed to climate stress have the least capacity to recover from it.

The Green Gentrification Problem

Most biophilic design discourse avoids a critical tension: green infrastructure investments can trigger displacement. Without deliberate protections, neighbourhoods receiving biophilic improvements often see rapid increases in property values and rents, pushing out the very residents who most needed the intervention.

This isn't theoretical. It's documented in cities across North America and Europe. The pattern is consistent: announce major park development or greening initiative in a disinvested neighbourhood, watch property speculation accelerate, observe demographic change within 5-7 years. The intended beneficiaries – long-term residents with high climate vulnerability – end up displaced to neighbourhoods with even less green infrastructure.

Louisville, Kentucky's experience with the Trust for Public Land's "10-Minute Walk" campaign illustrates both the challenge and potential solutions. The project prioritised park development in underserved neighbourhoods with high heat vulnerability. Targeted greening (2015-2023) correlated with 4-7°F surface temperature reductions and community-reported improvements in mental well-being.

Critically, the project incorporated anti-displacement protections from inception: community land trust models for adjacent residential development, participatory governance structures giving residents decision-making authority, and commitments to affordable housing preservation. Early indicators through 2023 (eight years post-implementation) suggest these protections are helping stabilise the community. The critical test will come in years 10-15 post-implementation, when property value pressures typically peak in similar interventions and displacement risks accelerate.

The point: equity requires more than directing investments toward marginalised communities. It requires structural protections ensuring those communities can remain and benefit from improvements. Otherwise, we're just using public resources to subsidise displacement and gentrification.

Social Infrastructure Amplifies Biophilic Benefits

Community-led and culturally responsive design processes aren't just ethical imperatives – they produce better outcomes. Projects shaped by local knowledge and priorities are more likely to be used, maintained, and experienced as safe. They reinforce social cohesion and psychological well-being in ways that top-down expert-led interventions often miss (Anguelovski et al., 2022).

Social connection itself functions as a stress buffer, amplifying biophilic environments' restorative effects through what amounts to collective regulation. When people experience green spaces together – through organised activities, informal gathering, or simply regular co-presence – the mental health benefits exceed what individual nature exposure would predict. The mechanism appears to involve both direct social support and the creation of shared place attachment, both of which protect against environmental stress.

This suggests that biophilic design should be considered alongside community organizing, cultural programming, and social infrastructure development, not as a replacement for those investments but as a complement.

Policy and Systemic Change

Climate-responsive biophilic design scales through policy frameworks embedding recovery, cooling, and ecological function into everyday infrastructure as default condition rather than exceptional achievement.

Zoning determines climate exposure before architects or designers get involved. Setback requirements, parking minimums, impervious surface allowances, and floor area ratios directly influence tree canopy potential, surface temperatures, and opportunities for shaded movement. Most existing zoning prioritises vehicular access and parking over thermal comfort or ecological function.

Jurisdictions that have reformed zoning to reduce parking minimums, incentivise green coverage, and require permeable surfaces show lower surface temperatures and improved thermal comfort (Hoffman et al., 2020). Portland's Green Street Policy, which requires stormwater management through vegetated infrastructure on all new streets, has yielded 25% tree canopy increases in targeted districts over 15 years. The policy works because it changed the baseline: green infrastructure became default rather than an upgrade requiring special justification.

Building Standards and Certification

Health-oriented frameworks like WELL and Fitwel incorporate criteria related to thermal comfort, air quality, daylight, and nature access, explicitly linking environmental conditions to cognitive performance and mental well-being (International WELL Building Institute, 2023; Center for Active Design, 2023). While voluntary, these systems increasingly influence public procurement and institutional development, creating market pressure toward climate-responsive biophilic features.

The limitation is that certification systems reward individual buildings rather than neighbourhood-scale systems. A WELL-certified building in a heat-island neighbourhood with no tree canopy or shaded pedestrian infrastructure provides limited refuge – the stress accumulates during daily movement. The next generation of standards needs to address this mismatch.

Heat action plans and climate resilience funding programs recognise mental health dimensions of heat exposure (Thompson et al., 2023). The opportunity is integration: shaded outdoor waiting areas at health centres, green buffers around public facilities, vegetated transit corridors. These extend public health interventions beyond emergency response into environmental regulation supporting daily mental health.

Most streets are designed exclusively for moving vehicles efficiently. This produces environments maximizing heat exposure and sensory stress while minimizing opportunities for thermal refuge. Reimagining streets as multimodal corridors that also function as thermal infrastructure would transform daily environmental experience for millions of urban residents.

Tree-lined corridors, shaded sidewalks, and green medians reduce thermal load while improving walkability and supporting calmer movement patterns (Aram et al., 2019). The design question: what if we treated streets as climate adaptation infrastructure with the same seriousness currently reserved for flood control or utilities?



Economic Arguments That Actually Work

Policy advocacy often emphasises environmental or social benefits, which can be compelling to some audiences but rarely drives budget allocations. Economic arguments tend to work better.

Every dollar invested in urban tree canopy yields $1.50-$3.00 in combined measurable benefits: energy savings from reduced cooling demand, healthcare cost reductions from improved air quality and physical activity, property value increases, stormwater management cost avoidance (Aram et al., 2019). Mental health benefits are harder to monetise but represent significant avoided costs in emergency department utilization, psychiatric medication, and long-term disability.

Framing biophilic climate adaptation as cost-effective risk reduction rather than amenity enhancement shifts the policy conversation. It's not about making nice spaces – it's about reducing liability and avoiding costs that will otherwise hit public health and emergency systems.

What This Means for Practice

The research synthesised here challenges several assumptions common in both climate adaptation planning and biophilic design practice.

First, climate adaptation is not primarily an engineering problem. It's a neurological problem that requires design responses addressing how human nervous systems respond to chronic environmental stress. Technical cooling strategies that ignore perceptual, cognitive, and emotional dimensions will underperform compared to integrated approaches.

Second, biophilic design is not aesthetic enhancement. Under climate stress, it functions as mental health infrastructure – as essential as cooling centers but operating preventively rather than reactively. This reframing changes how we prioritise investments, measure outcomes, and justify costs.

Third, equity cannot be an afterthought. Without structural protections against displacement, green infrastructure investments in marginalised communities often accelerate gentrification, ultimately harming the intended beneficiaries. Anti-displacement measures must be integrated from project inception, not layered on after property values start rising.

Fourth, networks matter more than nodes. Isolated refuges don't address cumulative stress burden across daily movement patterns. Continuous networks enabling graduated thermal transitions produce better mental health outcomes than equivalent green space concentrated in discrete locations.

Fifth, layer interventions across temporal scales. Projects combining immediate interventions (built shade, water features) with long-term strategies (tree canopy development) deliver both rapid relief and compounding benefits. Communities cannot wait 15-20 years for mature ecological function while suffering through intensifying heat waves.

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Measurement and Evaluation

Cities implementing biophilic climate adaptation need evaluation frameworks addressing both environmental performance and mental health outcomes. Surface and ambient temperature monitoring documents physical cooling effects. Utilization data (pedestrian counts, time spent in spaces, activity patterns) indicates whether interventions support behaviour change. Health outcome tracking (emergency department visits, psychiatric medication prescriptions, self-reported stress and well-being) measures mental health impacts.

The challenge is temporal: environmental changes occur relatively quickly (months to years) while mental health outcomes may take longer to manifest and attribute (years to decades). This creates political pressure to declare success prematurely based on environmental metrics alone, potentially missing longer-term mental health effects – both positive and negative.

Rigorous evaluation requires sustained monitoring over 5-10 year periods, with particular attention to equity dimensions. Who benefits? Who is excluded or displaced? Do improvements reach populations with highest climate vulnerability? Without this sustained attention, we risk replicating existing inequities under the banner of climate adaptation.

Research Priorities for the Next Decade

Three research directions would substantially advance both theory and practice.

First, controlled longitudinal studies examining mental health trajectories in matched communities with and without biophilic interventions. Most current evidence is correlational or short-term. Multi-year tracking would clarify whether benefits persist, identify threshold effects (minimum green space coverage for measurable impact), and reveal unintended consequences. This research is expensive and methodologically complex, but it's what funders and policymakers need to justify sustained investment.

Next, mechanistic studies isolating specific design variables and their neurological effects. Does shade matter more than cooling? What vegetation densities produce measurable stress reduction? How do different spatial configurations influence cognitive load? Understanding these relationships would enable precision design rather than generic "add more green" recommendations. The challenge is that mechanistic clarity often requires controlled settings that sacrifice ecological validity. Creative methodological approaches combining lab studies, field experiments, and natural experiments will be essential.



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Finally, implementation science examining how biophilic policies translate to on-the-ground outcomes. Most studies examine well-maintained demonstration projects. What happens when maintenance budgets shrink? How do communities sustain biophilic infrastructure through political transitions? What governance models support long-term ecological function? Implementation science rarely gets the attention that intervention research does, but it determines whether good ideas actually scale.

Additional research directions that would strengthen the evidence base include: better cultural and geographic diversity in study populations (the vast majority of current research occurs in Western, high-income contexts); research on indoor-outdoor integration and how different combinations optimise recovery; economic valuation studies providing comprehensive cost-benefit analyses including mental health outcomes; and climate scenario modelling examining how interventions perform under different warming projections.

That is, we need honest engagement with what we don't know and what may be unknowable. Climate scenarios project increasingly extreme conditions beyond anything in the historical record. Will biophilic strategies that work under current conditions continue working under 3-4°C warming? At what point do environmental stressors overwhelm biophilic buffers? These aren't comfortable questions but avoiding them doesn't make the uncertainty disappear.

The Path Forward

The science is clear enough to act on: climate stress is neurological, the built environment either amplifies or buffers that stress, and biophilic design provides evidence-based strategies for supporting mental health under chronic environmental instability. What remains uncertain is whether we'll treat this insight as actionable or continue framing climate adaptation primarily as engineering and ignoring the neurological dimension until mental health systems are overwhelmed.

The stakes are higher than many designers realise. Every degree of temperature rise erodes cognitive and emotional capacity. Every additional stressor reduces the reserves people need to adapt. Environments that compound rather than buffer climate stress don't just make people uncomfortable – they undermine the human capacities (attention, empathy, judgment, cooperation) that climate resilience depends on.

Recovery isn't optional. It's a prerequisite for sustained adaptation. As climate change reshapes our environments, the challenge isn't only building structures that physically endure. It's designing systems that allow minds and bodies to recover from repeated exposure to conditions that are, increasingly, beyond what human nervous systems evolved to handle.

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References

Anguelovski, I., Connolly, J. J. T., Pearsall, H., Shokry, G., Checker, M., Maantay, J., Gould, K. A., Lewis, T., Maroko, A., & Roberts, J. T. (2022). Green gentrification and health equity: The unintended consequences of urban greening. *Annual Review of Public Health, 43*, 121–140. https://doi.org/10.1146/annurev-publhealth-052020-110253

Aram, F., Solgi, E., García, E. H., Mosavi, A., & Band, S. S. (2019). Urban green space cooling effect in cities. *International Journal of Environmental Research and Public Health, 16*(24), 4797. https://doi.org/10.3390/ijerph16244797

Baecker, L., Iyengar, U., Del Piccolo, M. C., & Mechelli, A. (2025). Impacts of extreme heat on mental health: Systematic review and qualitative investigation of the underpinning mechanisms. *The Journal of Climate Change and Health, 22*, 100446. https://doi.org/10.1016/j.joclim.2025.100446

Center for Active Design. (2023). *Fitwel certification system: Health-promoting design strategies*. https://centerforactivedesign.org/fitwel

Clayton, S., Manning, C. M., Krygsman, K., & Speiser, M. (2017). *Mental health and our changing climate: Impacts, implications, and guidance*. American Psychological Association & ecoAmerica. https://www.apa.org/news/press/releases/2017/03/mental-health-climate.pdf

Evans, G. W. (2003). The built environment and mental health. *Journal of Urban Health, 80*(4), 536–555. https://doi.org/10.1093/jurban/jtg063

Hoffman, J. S., Shandas, V., & Pendleton, N. (2020). The effects of historical housing policies on resident exposure to intra-urban heat. *Climate, 8*(1), 12. https://doi.org/10.3390/cli8010012

Hsu, A., Sheriff, G., Chakraborty, T., & Manya, D. (2021). Disproportionate exposure to urban heat island intensity across major U.S. cities. *Nature Communications, 12*, 2721. https://doi.org/10.1038/s41467-021-22799-5

International WELL Building Institute. (2023). *WELL Building Standard v2*. https://www.wellcertified.com

Kaplan, R., & Kaplan, S. (1989). *The experience of nature: A psychological perspective*. Cambridge University Press. https://doi.org/10.1017/CBO9780511623390

Koc, C. B., Osmond, P., Peters, A., & Irger, M. (2018). Evaluating the cooling effects of green infrastructure: A systematic review. *Building and Environment, 134*, 111–124. https://doi.org/10.1016/j.buildenv.2018.02.030

McEwen, B. S., & Akil, H. (2020). Revisiting the stress concept: Implications for affective disorders. *Journal of Neuroscience, 40*(1), 12–21. https://doi.org/10.1523/JNEUROSCI.0733-19.2019

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Peters, A., McEwen, B. S., & Friston, K. (2017). Uncertainty and stress: Why it causes diseases and how it is mastered by the brain. *Progress in Neurobiology, 156*, 164–188. https://doi.org/10.1016/j.pneurobio.2017.05.004

Pihkala, P. (2020). Anxiety and the ecological crisis: An analysis of eco-anxiety and climate anxiety. *Sustainability, 12*(19), 7836. https://doi.org/10.3390/su12197836

Rony, M. K. K., Akter, S., & Moniruzzaman, M. (2023). High ambient temperature and mental health outcomes: A systematic review. *Environmental Research, 216*, 114445. https://doi.org/10.1016/j.envres.2022.114445

Thompson, R., Lawrance, E. L., Roberts, L. F., Grailey, K., Ashrafian, H., Maheswaran, H., Toledano, M. B., & Darzi, A. (2023). Ambient temperature and mental health: A systematic review and meta-analysis. *The Lancet Planetary Health, 7*(7), e580–e589. https://doi.org/10.1016/S2542-5196(23)00104-3

Twohig-Bennett, C., & Jones, A. (2018). The health benefits of the great outdoors: A systematic review and meta-analysis of greenspace exposure. *Environmental Research, 166*, 628–637. https://doi.org/10.1016/j.envres.2018.06.030

Ulrich, R. S., Simons, R. F., Losito, B. D., Fiorito, E., Miles, M. A., & Zelson, M. (1991). Stress recovery during exposure to natural and urban environments. *Journal of Environmental Psychology, 11*(3), 201–230. https://doi.org/10.1016/S0272-4944(05)80184-7

Yao, W., Zhang, X., Gong, P., Lin, Y., & Chen, Y. (2021). Effects of exposure to the natural environment on stress-related physiological outcomes: A meta-analysis. *Environmental Research, 195*, 110821. https://doi.org/10.1016/j.envres.2021.110821

